

Rung 0 — Orientation (read this first, it's 2 minutes)

Welcome. This track teaches the math and formal verification underneath the guarded, certified kernels on this site — the *why it's provably true* beneath the *what we built*. It's a study ladder, and this page is the "you are here" before you take the first step.

Who this is for

You, if you have roughly **Calc I/II**, know **binary and fixed-point numbers**, and have seen a **signal on a scope**. That's the launchpad. You do **not** need real analysis, logic, control theory, or any proof experience — those get introduced *just in time*, motivated by a real thing on the bench, never "abstract first, hope you see why later."

If some of it never got taught to you at the AS/BS-EE level — that's the point. It wasn't taught to the person building this track either. We're learning it together, out loud.

The one reframe that makes it feel possible

The scary-sounding stuff (formal verification, real analysis, o-minimality) is real, but the *first* genuinely new skill on the whole climb **isn't a math topic** — it's **learning to read a proof**. An engineering degree teaches you to *compute* answers; it never teaches you to *argue* that something is true for **every** input. That one skill is the gate (Rung 1.5), and it's smaller than it looks. Everything after leans on it.

How each rung works — the five beats

Every lesson has the same shape, so you always know where you are:

1. **The bench** — a real artifact (a scope trace, an evidence packet) and the question it raises.
2. **The one new idea** — the smallest new piece of math, in plain language.
3. **Do it by hand** — you derive the bound yourself, with a pen.
4. **Make it formal** — the matching Lean snippet, so you see what the machine checks.
5. **Watch it hold** — the real hardware capture that confirms it.

Then a short **Exercises & checkpoint** so you can tell whether it stuck before moving on.

The climb, at a glance

- **Rungs 1 → 1.5 → 2** — the first arc. Reachable with your EE math plus a little proof practice. By the end you'll have re-derived an entire machine-checked certificate by hand.
- **Rung 3** — how errors stack through a whole computation.

- **Rung 4** — Lean, the proof-checking machine. **The first real cliff** — a new tool with its own curve. Budget weeks, lean on the Natural Number Game. The track is honest about this.
- **Rung 5** — safety that holds *forever* (control loops, Lyapunov).
- **Rung 6** — the summit: what can even *be* certified. Research-level; a map of the peak, not a weekend hike.

How to actually use it

1. Read a rung slowly. Keep a pen out for beat 3.
2. Do the exercises. This is a skill — you learn it by doing, like soldering.
3. When a sentence moves too fast or loses you, **note which one**. That feedback is how the lessons get retuned to fit how *you* learn. Getting stuck isn't failing; it's data.
4. Don't rush the cliffs (Rungs 4 and 6). Everything below them is solid ground — enjoy it.

That's the whole map. There's no deadline and no grade — just a corner of math worth being fascinated by, made honest by a proof and confirmed on a real bench.

→ **Take the first step:** Rung 1 — How wrong is it, provably?